Paired-input constraints on quantum channels

Research direction and responsibility · Kirtus Justus. mathematical and computational assistance · ChatGPT (OpenAI). 7 November 2026, Port Coquitlam, British Columbia, Canada.

The supplied population values imply $\mathrm{Tr}(C^2) \le 0.742865465969$ for the normalized Choi matrix of the specified trace-preserving channel. The calculation also predicts a bound on the unmeasured joint output for a neighboring input. Both conclusions require the same channel to describe the two preparations.

Let w be the normalized Choi population of the four-dimensional space of operators $A\otimes I_2$, with normalized vectorization $|K\rangle=\mathrm{vec}(K)/2$. On input $|00\rangle$, a is the largest eigenvalue of the first output marginal, and β is the probability of finding the second qubit in state 1. At the supplied model point $w\ge0.99$ and $a\approx0.529472934504$. These are model populations. No device readings or measured input-transfer errors were supplied.

Quantity	Conditional result at the model point
Second-qubit flip on $\|00\rangle$	$\beta \le 0.039797974645$ (outward).
Largest joint-output eigenvalue on $\|01\rangle$	At most 0.824617080524 (outward). This prediction can be held out.
Choi purity using both input constraints	At most 0.742865465969 (outward). Global optimality is open.
Choi half-trace distance to any unitary	At least 0.173002789690 (outward). Exact local-data result on page 3.

Input values and the single-input result

Use match population $s=131/320$, reference response $h=99/100$, ordinary probability $q=81711338569043/224995905216000$, and orthogonal-projector population f. The detector ray is $\sqrt{h}|I_4\rangle+\sqrt{(1-h)}|iZ\otimes I\rangle$. The additional outcome projects onto $|iX\otimes I\rangle$. Positivity gives $w\ge s+f+(\sqrt{q}-\sqrt{(hs)})^2/(1-h)$. At the specified 1% residual point the right side is 0.99.

For local readings alone put $m=\min(1,a+\min(\beta,1-\beta))$. The sharp purity upper endpoint is $H(m)=[9+6m+m^2+(1-m)^2]/16$. The lower endpoint and unitary criterion are supported. Combining the sharp flip relation on page 2 with H gives the valid outward ceiling 0.807826398288. The tighter bound in the table uses another consequence of w.

What the local Choi population constrains

Write each Kraus matrix K_j in second-bit blocks A_j,B_j,C_j,D_j. Orthogonal projection onto operators of the form $V\otimes I_2$ gives $4(1-w)=\Sigma_j[\tfrac{1}{2}\|A_j-D_j\|F^2+\|B_j\|F^2+\|C_j\|F^2]$. Let β be the flip probability on $|00\rangle$ and α the reverse flip on $|01\rangle$. Keeping the first columns and applying the reverse triangle inequality to the concatenated diagonal blocks bounds the right side below by $\beta+\alpha+\tfrac{1}{2}(\sqrt{(1-\alpha)}-\sqrt{(1-\beta)})^2$. This expression increases in α, giving $4(1-w)\geq\beta+\tfrac{1}{2}(1-\sqrt{(1-\beta)})^2$.

Equality for each β is achieved by $\mathrm{diag}(\sqrt{(1-\beta)},1,1,1)$ and $\sqrt{\beta}\,|01\rangle\langle 00|$ as Kraus operators. This proves sharpness when w and β are the only constraints. At $w\geq 0.99$, inversion yields $\beta\leq 1-(\sqrt{3.92}-1)^2$.

Comparison of two inputs

Let V be the Stinespring isometry from these Kraus matrices. In a common environment compare $\psi=V|00\rangle$ with $\varphi=(X_2\otimes I_{env})V|01\rangle$, where X_2 flips the second output bit. Their squared difference is $\Sigma_j[\|(A_j-D_j)|0\rangle\|^2+\|(C_j-B_j)|0\rangle\|^2]\leq 8(1-w)$. Consequently $\mathrm{Re}\langle\psi|\varphi\rangle\geq 4w-3\geq c=0.96$. X_2 is a mathematical change of output coordinates in this argument.

The largest eigenvalue of the joint output on $|00\rangle$ is at most $m_0=a+\beta_{max}\approx 0.569270909148$. Indeed its two positive diagonal blocks have operator norms at most a and β, and positivity bounds the off-diagonal contribution by their geometric mean. The same eigenvalue cap holds in the environment, because ψ is pure.

For any environmental ray let p_0,p_1 be its probabilities in ψ and φ. Then $p_0\leq m_0$ and $c\leq\sqrt{(p_0p_1)}+\sqrt{[(1-p_0)(1-p_1)]}$. Writing each square-root pair as a unit vector gives $p_1\leq m_1=(c\sqrt{m_0}+\sqrt{[(1-c^2)(1-m_0)]})^2$ since $m_0<c^2$. This is uniform over rays. The environment and joint output on $|01\rangle$ thus have largest eigenvalue at most $m_1\approx 0.824617080524$.

> The environment on a maximally mixed input is the average of its four basis-input states and has the nonzero spectrum of C. Therefore $\lambda_{max}(C)\leq t=(m_0+m_1+2)/4$. Since $t\geq\tfrac{1}{2}$, $(t,1-t,0,\ldots)$ majorizes the spectrum of C and $\mathrm{Tr}(C^2)\leq t^2+(1-t)^2<0.742865465969$.

A sharp distance from the local readings alone

For the specified first marginal and second outcome β, the minimum of $\frac{1}{2}\|C-P_u\|_1$ over compatible trace-preserving channels C and unitary Choi states P_u is the following function of $m=\min(1,a+\min(\beta,1-\beta))$.

> $D_{\min}(m) = [1-m + \sqrt{((1-m)^2+12(1-\sqrt{m})^2)}]/8$. The formula also applies when the complete joint output Ω is fixed, with $m=\lambda_{\max}(\Omega)$. It is a normalized Choi-state distance.

Lower bound

Fix a candidate unitary U. Its Choi vector in the input basis is $(g+\sqrt{3}\,v)/2$, where $g=U|0\rangle\otimes|0\rangle$ and $v=\Sigma_{j=1}^{3} U|j\rangle\otimes|j\rangle/\sqrt{3}$. Here $|0\rangle$ denotes the tested four-dimensional input $|00\rangle$. The g diagonal of C is at most m/4 from the output bound. The v diagonal is at most ¾ from trace preservation on the three-dimensional complementary input subspace. Positivity bounds the magnitude of the cross entry by $\sqrt{(3m)/4}$.

On any unit vector $xg+yv$ with x,y real nonnegative, the expectation of P_u-C is therefore bounded below by the Rayleigh quotient of the matrix with entries $[(1-m)/4, \sqrt{3}(1-\sqrt{m})/4; \sqrt{3}(1-\sqrt{m})/4, 0]$. Its top eigenvector can be chosen nonnegative and its top eigenvalue is $D_{\min}(m)$. Half-trace distance bounds the expectation of a rank-one test for a traceless difference. This proves the lower bound for every compatible C,U.

Attainment

Choose the joint output attaining eigenvalue m. For local data with $a+b<1$ and $b=\min(\beta,1-\beta)$, one such output is $|z\rangle\langle z|+(1-a-b)|10\rangle\langle 10|$, where $z=\sqrt{a}|00\rangle+\sqrt{b}|11\rangle$ in the corresponding bases. For $a+b\geq 1$ use a pure joint output with the specified marginals and second probability. For a fixed joint output take its eigenvectors u_j with eigenvalues λ_j in descending order.

Set $U|j\rangle=u_j$. Use Kraus matrices $K_0=U\,\mathrm{diag}(\sqrt{m},1,1,1)$ and $K_j=\sqrt{\lambda_j}\,|u_j\rangle\langle 0|$ for j=1,2,3. The operators preserve trace and give the chosen joint output. The residual Choi components are perpendicular to g,v. Thus P_u-C has only one positive eigenvalue when $m<1$, exactly the one computed above. At $m=1$ take $C=P_u$.

The model gives an upper limit on m. $D_{\min}$ decreases with m. Substitution gives distance at least 0.173002789690 to every unitary Choi state. With the weaker purity ceiling 0.807905661518 alone, the sixteen-dimensional spectral argument gives approximately 0.101546767054 instead. Both apply to a mathematical observation class.